Quang-Tuan DANG

02/10/1995

Yau Mathematical Sciences Center, Tsinghua Univ.
China, 100084, Beijing, Haidian District
dangquangtuan10@gmail.com
quangtuandang.github.io/Webpage/

Research Interests

Complex and Differential Geometry, geometric analysis and PDEs, more precisely, complex de Monge-Ampère equations, canonical metrics, and geometric flows

Employment

09/2027– **Tenure-Track Assistant Professor**, *Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), Fudan University, Shanghai*

2025–2027 **Postdoc**, *Yau Mathematical Sciences Center, Tsinghua University, Beijing, mentored by Shing-Tung Yau*

2022–2025 **Postdoc**, *International Centre for Theoretical Physics (ICTP), Italy, mentored by Claudio Arezzo*

Education

2019-2022 **Doctorat de Mathématiques Fondamentales**, *LMO, Université Paris-Saclay*
Title: Pluripotential Monge-Ampère flows
Advisors: Hoang-Chinh LU & Vincent GUEDJ

2017–2019 **Master de Recherche en Mathématiques Fondamentales**, *Université Paris-Saclay*

2013–2017 **Bachelor of Education**, *Hanoi National University of Education*
Excellent, Mathematics

Publications et preprints

2026 *Uniform estimates for complex Monge-Ampère equations II: Hermitian manifolds* (with Lei Zhang and Bin Zhou), arXiv:2610.39259

Uniqueness and Stability of Monge–Ampère Potentials in Big Cohomology Classes (with Lei Zhang and Bin Zhou), arXiv:2609.00810

Uniform estimates for complex Monge-Ampère equations: big cohomology classes (with Lei Zhang and Bin Zhou), arXiv:2608.19600

Regularity of solutions to Monge–Ampère equations on compact Hermitian manifolds, **J. London Math. Soc., 114(4):e70746, 2026.**

2025 *Modulus of continuity of Monge–Ampère potentials in big cohomology classes* (with Hoang-Son Do, Hoang Hiep Pham), arXiv:2504.15763, accepted to Indiana Univ. Math. J.

Singularities vs non-pluripolar Monge–Ampère masses (with Hoang-Son Do, Hoang Hiep Pham), **Math. Z. 311, 47 (2025).**

2024 *An iterative construction of complete Kähler-Einstein metrics* (with Tat Dat Tô), **Publ. Res. Inst. Math. Sci. 62 (2026), no. 4, pp. 629–661**

Hermitian null loci, **Trans. Amer. Math. Soc.**

2023 *Kähler–Einstein metrics on quasi-projective manifolds* (with Duc-Viet VU), **Math. Ann. 392, 4071–4103 (2025)**

Singularities of the Chern-Ricci flow, **Anal. PDE Vol. 19 (2026), No. 3, 449–483**

Continuity of Monge-Ampère Potentials with Prescribed Singularities, **J. Geom. Anal. 33, 318 (2023).**

2022 *Pluripotential Chern-Ricci Flows*, **Indiana Univ. Math. J. 73 No. 4 (2024), 1401–1441**

2021 *Pluripotential Monge-Ampère flows in big cohomology classes*, **J. Funct. Anal. 282 (2022), no. 6, Paper No. 109373, 65 pp.**

Continuity of Monge-Ampère Potentials in Big Cohomology Classes, **Int. Math. Res. Not. IMRN 2022, no. 14, 11180–11201.**

Talks

Oct. 2026 Analysis PDE Seminar, BIT, Beijing

June 2026 Geometric Analysis Seminar, Chinese Academy of Sciences, Beijing

May 2026 Seminar, Institute for Advanced Study in Mathematics of HIT, Heilongjiang, China

April 2026 Geometric Analysis Seminar, Peking University, Beijing, China

April 2026 Seminar, Westlake University, Hangzhou, China

Jan. 2026 IMFP-IGP Seminar, USTC Shanghai, China

Nov. 2025 Seminars, Beijing Normal University, Beijing, China

Nov. 2025 USTC complex geometry seminar, Hefei, China

June 2025 Workshop: Géométrique Algébrique et Géométrie Complexe, M'bour, Senegal

Jan. 2025 Analysis Seminar, Institute of Mathematics, VAST, Hanoi, Vietnam

Jan. 2025 Geometry Seminar, Hanoi University of Sciences, Hanoi

Dec. 2024 Séminaire Géométrie et Topologie, Brest

Dec. 2024 Séminaire d'Analyse et Géométrie, Jussieu

Oct. 2024 Seminari di Geometria Differenziale e Analisi Complessa, Florence, Italy

Sep. 2024 Conference: Progressi recenti in geometria reale e complessa XIII, 24-27 Sep, Levico Terme (Italy)

Aug. 2024 Gothenburg Complex Geometry, 26-30 Aug., Gothenburg (Sweden)

Sep. 2023 Workshop: Geometric flows on complex manifolds, 11-15 Sep, Cortona (Italy)

June 2023 ICTP-IGAP seminar, Trieste, Italy

Feb. 2023 Complex Geometry seminar, Cologne, Germany

Jan. 2023 Analysis Seminar, Institute of Mathematics, VAST, Hanoi, Vietnam

Nov. 2022 ICTP-IGAP seminar, Trieste, Italy

June 2022 International Workshop, AIAS, Aarhus, Denmark

June 2022 MARGE Workshop, IMAG, Montpellier, France

April 2022 IGA seminar, University of Maryland, New York, United States

Nov. 2021 Informal Workshop: Balanced metrics and Monge-Ampere masses, Parma, Italy

Oct. 2021 Séminaire de Géométrie Complexe, IMT, Toulouse

June 2021 Journée doctorale: Analyse, LMO, Orsay, France

Service

Referee for International Mathematics Research Notices, Acta Mathematica Vietnamica, Complex Manifolds, Complex Analysis and Operator Theory, Analysis & PDE, Indiana University Mathematics Journal, Pacific Journal of Mathematics

Reviewer for MathSciNet

Teaching

2026-2027 Differential Geometry (TA)
2025-2026 Differential Geometry (TA), Calculus (TA), Complex Geometry (TA)
2024-2025 Topology (12h TA), Partial Differential Equations (9h TA)
2023-2024 Partial Differential Equations (9h TA)
2022-2023 Partial Differential Equations (9h TA)
2021-2022 Analyse 3 en L2 (TD 36h), Mathématiques en L1 (32h TA), Python (6h TA)
2020-2021 Mathématiques en L1 (62h TA)

Languages

Vietnamese (mother tongue), English (fluent), French (fluent), Italian (intermediate), Chinese (intermediate)

Informatic

LaTeX, langue html, Python, Microsoft Office